Neutrino Telescopes at Sea: ANTARES and KM3NeT

Research Report by Aryan (20231053)

Abstract

The topic of this review is how neutrino telescopes work, as we know that most of the signal from outer space we get is in the form of high energy particles(neutrinos). Because neutrinos possess an exceptionally low interaction cross-section. We will examine the evolution of deep-sea Cherenkov neutrino observatories in the Mediterranean Sea, focusing on the pioneering ANTARES detector and its next-generation successor, KM3NeT (comprising the ARCA and ORCA arrays). Furthermore, we evaluate the harsh environmental challenges inherent to abyssal operation including natural ^{40}K radioactive decay, marine bioluminescence, and hydrodynamic drag. We detail the technological innovations driven by deep-sea physics, notably sub-nanosecond White Rabbit timing synchronization and subsea electro-optical infrastructure. Finally, using recent benchmark data and the landmark ultra-high-energy event `KM3-230213A`, we explore the emerging role of Mediterranean observatories within the Global Neutrino Network and evaluate future mitigation strategies for next-generation underwater astrophysics.

Keywords: Neutrino Astronomy, Cherenkov Radiation, ANTARES, KM3NeT, ARCA, ORCA, Deep-Sea Instrumentation, Multi-Messenger Astrophysics.

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1 Introduction and Astrophysical Motivation

For over a century, all the astrophysicist are completely depend exclusively on the electromagnetic spectrum. However, observing the universe at extreme energies ($E > 10\,\text{TeV}$) through photons encounters a fundamental physical barrier: high-energy gamma rays interact with background photon fields, such as the Cosmic Microwave Background (CMB) , via electron-positron pair production:

$$\gamma + \gamma_{\text{background}} \longrightarrow e^+ + e^- \tag{1}$$

This interaction made us unable to detect PeV gamma rays over cosmological distances. Similarly, charged cosmic rays (primarily relativistic protons and heavier nuclei) are deflected by galactic and intergalactic magnetic fields, so we have noi idea how and from where the particles came from.

Neutrinos offer an ideal solution to the cosmic-ray pointing problem. Having zero electric charge and interacting exclusively via the weak nuclear force and gravity, neutrinos travel undeflected directly from the core of cosmic accelerators(like: Blazers, Black Holes, Supenovas etc) to Earth. Detecting high-energy astrophysical neutrinos provides unambiguous evidence for hadronic acceleration processes inside sources such as active galactic nuclei (AGN), blazar jets, gamma-ray bursts (GRBs), and supernova remnants.

The first detectors of atmospheric neutrinos in 1965 were particle or nuclear physics types [Ach65a,Rei65b]. It was difficult to build huge detectors of this type. But, at the 1960 Rochester Conference, Moisei Markov had reported his brilliant and prescient idea of deep underwater neutrino detection: “we propose setting up apparatus in an underwater lake or deep in the Ocean to separate charge particle direction by Cherenkov radiation” [Mar60]. Indeed, water is very cheap and the target itself is not a limiting factor. Markov proposed that natural lakes, oceans, or deep glaciers could provide the immense target mass required to detect weakly interacting

neutrinos at manageable cost. Early explorations began with The first DUMAND workshop in 1975 [Kot75] chose Hawaii as a large volume of deep sea equipped with lines of photomultipliers. To observe neutrinos coming from astrophysical sources, so the detection principle focused on upward-traveling muons coming from high-energy (¿ 1 GeV) muon-neutrino interactions. The photomultipliers record arrival time and amplitude of Cherenkov light emitted by muons or particle cascades., the first prototypes of DUMAND failed, showing that ocean is also a hostile environment, and the funding agencies did not give a second chance. Developments in ice were more lucky and the first AMANDA prototype (80 optical modules) was deployed at the South Pole in 1993-1994. In Russia, the Baikal NT 200 was completed only in 1998, which culminated in the IceCube Neutrino Observatory [2].

While IceCube has achieved major milestones in glacial ice, a deep-sea detector in the Northern Hemisphere is indispensable. Due to the high flux of downward atmospheric muons produced in cosmic-ray air showers, neutrino telescopes use the Earth as a filter, looking primarily downward through the planet for upward-going neutrino-induced leptons. Consequently, a South Pole detector observes primarily the Northern celestial sky, while a Mediterranean observatory looks through the Earth directly toward the **Southern sky**, which hosts the **Galactic Center** and the majority of the Milky Way plane.

2 Physics of Underwater Neutrino Detection

2.1 Production of Cosmic and Atmospheric Neutrinos

Before examining how neutrinos interact in seawater, we must understand how they are produced. Unlike human-made particle accelerators that produce monoenergetic beams, astrophysical shocks produce power-law energy spectra through **first-order Fermi acceleration** (diffusive shock acceleration). The source injection spectrum of accelerated protons and ions follows:

$$\phi_p(E) \propto E^{-\alpha} \quad (\alpha \approx 2.0\text{–}2.2) \tag{2}$$

As cosmic rays propagate through the galactic magnetic field, their characteristic confinement and escape time decreases with energy, $\tau(E) \propto E^{-\delta}$ (with $\delta \approx 0.6$). Consequently, the cosmic-ray spectrum observed at Earth steepens to $\rho_{\rm CR}(E) \propto E^{-(2+\delta)} \approx E^{-2.7}$.

When these relativistic protons collide with ambient gas nuclei (pp interactions) or background radiation fields ($p\gamma$ interactions), they produce hadronic cascades dominated by charged pions ($\pi^{\pm}$) and kaons ($K^{\pm}$):

$$p + p \longrightarrow \pi^{\pm} + K^{\pm} + X \tag{3}$$

$$\pi^{+} \longrightarrow \mu^{+} + \nu_{\mu} \tag{4}$$

$$\hookrightarrow \mu^{+} \longrightarrow e^{+} + \nu_{e} + \bar{\nu}_{\mu} \tag{5}$$

This decay chain produces a characteristic flavor ratio at the source of $(\nu_e : \nu_\mu : \nu_\tau) \approx (1 : 2 : 0)$. Over cosmological propagation distances, standard vacuum neutrino oscillations average this ratio to an equal democratic mixture of approximately $(1 : 1 : 1)$ at Earth.

2.1.1 Three Distinct Neutrino Regimes

Based on these production environments, deep-sea observatories observe three distinct classes of neutrino spectra:

1. **Atmospheric Neutrinos:** Produced in Earth's upper atmosphere. They exhibit a soft spectrum ($\propto E^{-2.7}$, steepening to $E^{-3.7}$ above $\sim 100\,\text{GeV}$) and display a strong zenith-angle dependence because near-horizontal cosmic rays traverse a longer, more tenuous path where pions have more time to decay.

2. **Diffuse Galactic Neutrinos:** Produced by galactic cosmic rays interacting with the interstellar medium (gas and dust). Because interstellar gas densities are extremely low ($\sim 10^{10}\,\text{atoms/cm}^3$), secondary pions and muons always decay before interacting, producing an un-steepened spectrum following the galactic cosmic-ray index up to PeV energies.

3. **Astrophysical (Cosmic Accelerator) Neutrinos:** Produced directly at the acceleration sites (e.g., blazars, active galactic nuclei, gamma-ray bursts). Because they emerge directly from the accelerator without propagation steepening, they maintain the original hard Fermi shock spectrum ($\propto E^{-2.0\text{–}2.2}$), emerging prominently above the steeper atmospheric background at energies above tens of TeV.

2.2 Neutrino-Nucleon Interactions

Neutrinos cannot be detected directly; they are observed via relativistic charged leptons produced through deep inelastic scattering with atomic nuclei (N) in seawater or seabed rock. These interactions proceed via two distinct channels:

1. **Charged-Current (CC) Interaction:** Mediated by the exchange of a $W^\pm$ boson:

$$\nu_\ell + N \xrightarrow{W^\pm} \ell^- + X \quad (\ell = e, \mu, \tau) \tag{6}$$

   For muon neutrinos (ν_μ), the resulting relativistic muon can travel kilometers through water and bedrock, producing a distinctive, linear "track-like" topology.

2. **Neutral-Current (NC) Interaction:** Mediated by the neutral Z^0 boson:

$$\nu_\ell + N \xrightarrow{Z^0} \nu_\ell + X \tag{7}$$

   This process transfers energy to the hadronic remnant X, generating an energetic, localized hadronic shower ("cascade-like" topology). Electron neutrinos (ν_e) also produce electromagnetic cascades via CC interactions through $e^- \rightarrow$ bremsstrahlung.

2.3 Cherenkov Radiation in Seawater

When a charged particle travels through a dielectric medium at a speed v exceeding the phase velocity of light in that medium (c/n, where n is the refractive index), it polarizes local atoms along its trajectory. The coherent relaxation of these atoms emits a wave of shock in the sea that is electromagnetic radiation known as **Cherenkov radiation**.

The radiation is emitted along the surface of a cone characterized by a cherenkov angle θ_c:

$$\cos\theta_c = \frac{1}{\beta n} \tag{8}$$

where $\beta = v/c$. For ultra-relativistic muons ($\beta \approx 1$) in deep Mediterranean seawater ($n \approx 1.35$), the emission angle is approximately:

$$\theta_c \approx \arccos\left(\frac{1}{1.35}\right) \approx 42.1^\circ \tag{9}$$

The spectral distribution of emitted Cherenkov photons per unit path length dx is governed by the Frank-Tamm formula:

$$\frac{d^2N}{dx\,d\lambda} = \frac{2\pi\alpha}{\lambda^2}\left(1 - \frac{1}{\beta^2 n^2(\lambda)}\right) \tag{10}$$

where $\alpha \approx 1/137$ is the fine-structure constant. Because the photon yield is proportional to $1/\lambda^2$, the emission is concentrated in the ultraviolet and blue bands ($\lambda \approx 350\text{–}500\,\text{nm}$), where water transparency is maximal.

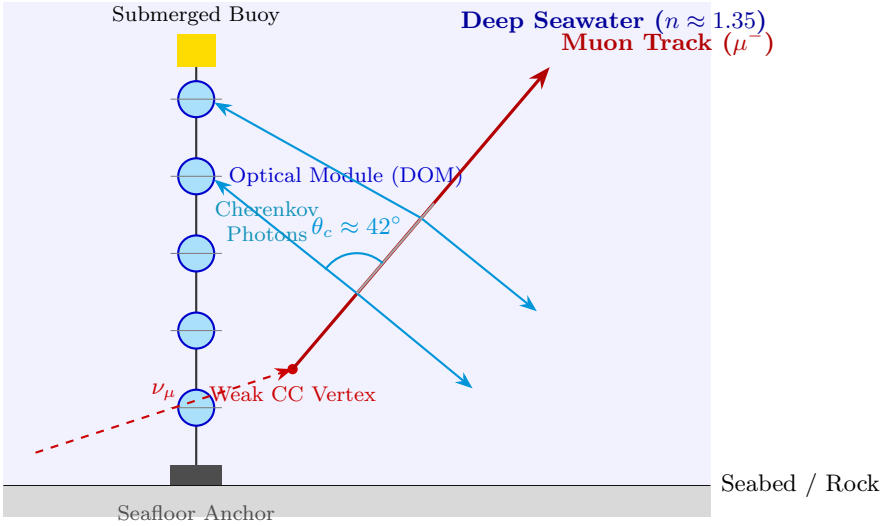


Figure 1: Schematic of deep-sea neutrino detection. An incoming ν_μ interacts in the seabed rock or water via charged-current interaction, producing a relativistic muon (μ^-). Moving faster than the local phase velocity of light, the muon emits a Cherenkov cone at $\theta_c \approx 42°$, registered by photomultiplier tubes housed along moored vertical strings.

Table 1: Architectural comparison between ANTARES and the KM3NeT detector arrays.

Parameter	**ANTARES**	**KM3NeT-ARCA**	**KM3NeT-ORCA**
Location	Toulon, France	Capo Passero, Italy	Toulon, France
Depth	2,475 m	3,500 m	2,450 m
Primary Physics	Galactic point sources	High-energy cosmic ν	Neutrino mass ordering
Energy Range	100 GeV – 100 TeV	1 TeV – 10 PeV	1 GeV – 100 GeV
Sensor Strings	12 mooring lines	2×115 strings	115 strings
Module Type	Storeys (3×10" PMTs)	Multi-PMT DOM	Multi-PMT DOM
PMTs per Module	3 large PMTs	31 small 3" PMTs	31 small 3" PMTs
Horizontal Spacing	$\sim$ 60 m	$\sim$ 90 m	$\sim$ 20 m
Vertical Spacing	14.5 m	36 m	9 m

2.4 Optical Properties: Seawater vs. Polar Ice

The physical medium critically influences detector reconstruction precision:

- **Glacial Ice (IceCube):** Features a very long absorption length ($\sim$ 100–200 m), allowing larger module spacing. However, residual air bubbles and dust layers produce a short scattering length ($\sim$ 20–50 m). Cherenkov Photons scatter repeatedly before detection, blurring their arrival times and limiting track angular resolution to approximately 0.5°–1.0°.
- **Deep Seawater (ANTARES / KM3NeT):** Liquid seawater exhibits minimal scattering, with an effective scattering length exceeding 100–200 m. Cherenkov Photons travel along unperturbed geometric paths, preserving sub-nanosecond arrival times. Consequently, underwater neutrino telescopes achieve exceptional pointing resolution ($< 0.1°$–$0.2°$ for muons above 10 TeV), making deep sea water optimal for neutrino point-source astronomy.

3 Detector Architecture: From ANTARES to KM3NeT

3.1 ANTARES: The Pioneer (2006–2022)

Constructed 40 km off the coast of Toulon at a depth of 2,475 m, the ANTARES observatory consisted of 12 flexible vertical lines anchored to the seabed and tensioned by submerged buoys. Each line held 25 storeys equipped with three 10-inch optical modules tilted 45° downward to minimize biofouling and optimize sensitivity to up-going Cherenkov light. Operating continuously for 16 years until its planned decommissioning in February 2022, ANTARES demonstrated that underwater optical arrays could operate reliably over decadal timescales.

Beyond proving operational viability, ANTARES confirmed atmospheric neutrino oscillations in deep sea water. Figure 2 shows the measured event rate as a function of the reconstructed $E/\cos\theta$ ratio (proportional to baseline over energy, L/E). The clear dip between 20–60 GeV rejected the un-oscillated hypothesis and agreed with world-best atmospheric mixing fits ($\nu_\mu \to \nu_\tau$).

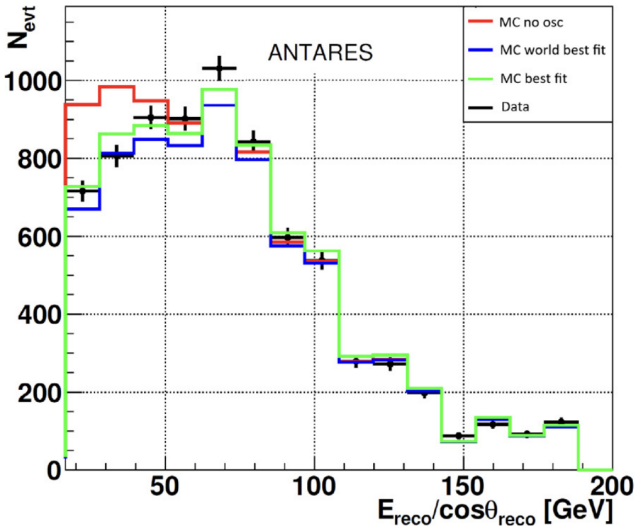


Figure 2: Reconstructed event rate as a function of $E_{\text{reco}}/\cos\theta_{\text{reco}}$ measured by ANTARES. Black points represent data; the red line shows the un-oscillated Monte Carlo prediction, while the green and blue curves show best fits including neutrino oscillations.

3.2 KM3NeT: The Multi-PMT Digital Optical Module (DOM)

Building upon ANTARES, the KM3NeT collaboration introduced a transformative advance in detection hardware: **the Digital Optical Module (DOM)**. Rather than housing a single large photomultiplier inside a glass sphere, the KM3NeT DOM incorporates **31 small 3-inch PMTs** inside a 17-inch pressure-resistant borosilicate sphere.

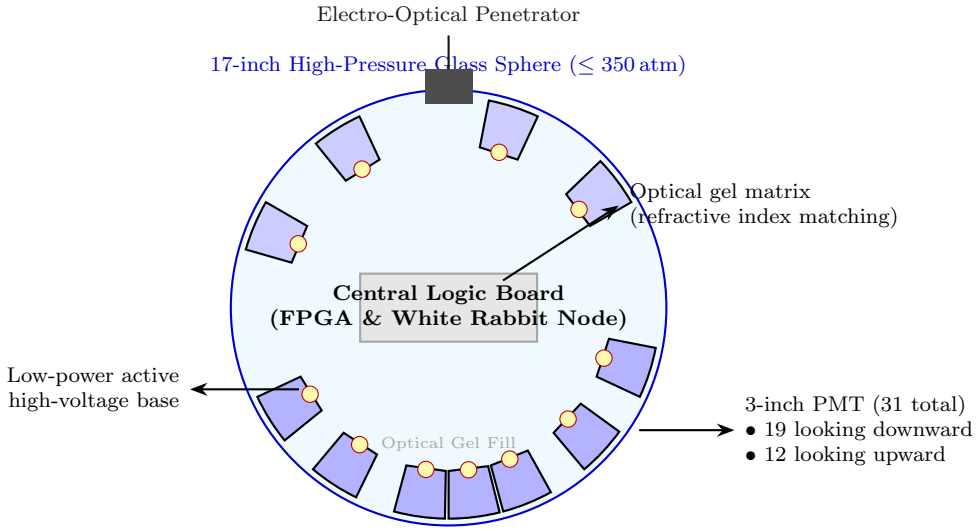


Figure 3: Cutaway schematic of the KM3NeT Digital Optical Module (DOM). A 17-inch pressure-resistant glass sphere houses 31 small 3-inch PMTs segmented into upper and lower hemispheres, embedded in optical gel and read out by an integrated FPGA board running the White Rabbit timing protocol.

This multi-PMT design yields key advantages:

1. **Directional Capability:** PMTs point in all directions (19 downward, 12 upward), providing nearly isotropic 4π coverage and segmenting the incoming Cherenkov wavefront.
2. **Local Background Suppression:** A single muon track induces simultaneous signals (coincidences) across multiple PMTs inside the same DOM. In contrast, random environmental optical noise rarely triggers multiple PMTs within a narrow nanosecond coincidence gate.
3. **Cost and Reliability:** Small 3-inch PMTs operate with low high-voltage bases and eliminate external magnetic shielding.

3.3 ARCA vs. ORCA: Two Telescopes, One Technology

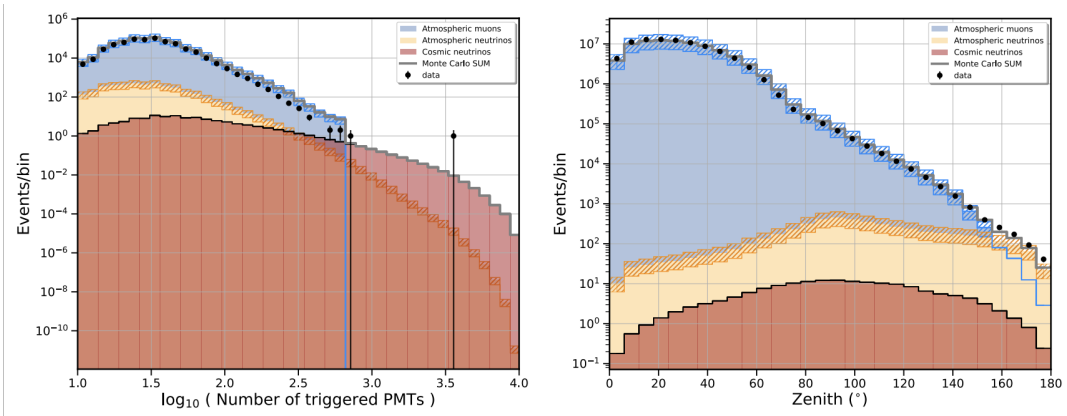
KM3NeT distributes this hardware across two sites tailored to distinct physics goals (Table 1):

- **KM3NeT-ARCA (Sicily, $3,500$ m):** Uses wide string separation (90 m) over a cubic-kilometer footprint to discover high-energy cosmic neutrino point sources in the TeV–PeV regime.
- **KM3NeT-ORCA (Toulon, $2,450$ m):** Employs a dense string configuration (20 m horizontal, 9 m vertical spacing) inside a few megatons of seawater. ORCA targets low-energy atmospheric oscillations (1–100 GeV) to determine the **Neutrino Mass Ordering (NMO)** - resolving whether the ν_3 mass eigenstate is heavier (normal) or lighter (inverted) than ν_1 and ν_2.

4 Operational Challenges in the Abyssal Environment

Operating at abyssal depths imposes severe physical and background constraints:

1. **Potassium-40 (^{40}K) Decay:** Natural seawater contains dissolved salts, including the radioactive isotope ^{40}K. Its decay channels (primarily β-decay: $^{40}\mathrm{K} \rightarrow {}^{40}\mathrm{Ca} + e^- +$



Extended Data Fig. 1 | Comparison between data and Monte Carlo simulations. The distribution of the $\mathcal{N}_{\text{trig}}^{\text{PMT}}$ observable is shown on the left and the reconstructed zenith angle on the right. Simulated atmospheric muons are shown in blue and atmospheric neutrinos are shown in yellow. A 40% systematic uncertainty on their normalization is also included in the histograms. The expected distribution for a cosmic neutrino flux described by the best fit from the IceCube Northern Sky Tracks[5] is shown for comparison in red.

Figure 4: Comparison between experimental data and Monte Carlo simulations for triggered PMTs (left) and reconstructed zenith angle (right). Atmospheric muons (blue) dominate the downward sky ($\theta < 90°$), whereas up-going tracks ($\theta > 90°$) reveal clean neutrino signals shielded by the Earth.

$\bar{\nu}_e$) generate relativistic electrons that emit Cherenkov light, creating a steady optical background of 20–40 kHz per PMT.

2. **Marine Bioluminescence:** Deep-sea marine organisms emit spontaneous flashes of light. While baseline bioluminescence contributes to ambient background rates, intense biological blooms can produce transient megahertz-level count rates, forcing temporary data gating.

3. **Cosmic-Ray Muon Background:** As illustrated in Figure 4, atmospheric muons arriving from above ($\theta < 90°$) outnumber genuine astrophysical neutrinos by more than six orders of magnitude. The telescope overcomes this noise by selecting up-going events ($\theta > 90°$), where thousands of kilometers of Earth's mantle filter out all downward-going muons.

4. **Currents and Mooring Dynamics:** Deep Mediterranean currents (up to 30 cm/s) cause vertical mooring lines to sway and tilt. To preserve angular reconstruction, high-frequency **Acoustic Positioning Systems (APS)** comprising seabed transceivers and hydrophones along the lines-triangulate every optical module's position in real time with an accuracy better than 10 cm.

5 Technological, Industrial, and Interdisciplinary Impact

5.1 Current Real-World Technological Spin-offs

Building deep-sea particle observatories produced a need for technical advancements fulfilled by subsea engineering, communications, and oceanography:

- **Sub-Nanosecond Clock Synchronization (White Rabbit):** Acc. to reports this is very important and great innovation that not only help these telescopes but also humankind. To reconstruct muon tracks, photon arrival times must be synchronized across kilometers of fiber optic cables to within < 1 ns. KM3NeT adapted and advanced the open-source **White Rabbit (WR)Ethernet protocol** developed at CERN. Today, White Rabbit is widely deployed in commercial telecommunications, 5G cellular networks, high-frequency financial trading, and smart power grids to ensure jitter-free, microsecond-level synchronization.

- **Deep-Sea Electro-Optical Infrastructure:** KM3NeT pioneered pressure-tolerant titanium junction boxes and wet-mateable electro-optical connectors operable by Remotely Operated Vehicles (ROVs) at pressures exceeding 350 atm. Controlling these devices not only reduces error in our experimentations but alos allow fluidity in experiments. These connectors are now widely used in commercial offshore energy and submarine communications cables.

- **Earth and Environmental Monitoring (EMSO Synergy):** The deep-sea infrastructure serves as a permanent benthic observatory connected to the European Multidisciplinary Seafloor and water-column Observatory (EMSO). High-resolution pressure gauges and seismometers provide real-time seabed geophysical data for tsunami and earthquake alert networks, while hydrophone arrays continuously monitor cetacean vocalizations (sperm whales and fin whales) to protect marine mammals from ship collisions in busy shipping lanes.

5.2 Future Industrial Potential: "If-Then" Projections

Continuing to advance deep-sea neutrino telescope technologies offers significant cross-sector industrial potential:

- **Autonomous Subsea Robotics:** *If* telescope R&D perfects autonomous inductive power transfer and optical data docking stations on the seafloor, *then* the offshore renewable energy and submarine pipeline industries can eliminate costly surface support vessels. Autonomous Underwater Vehicles (AUVs) will recharge and download maintenance data directly at seabed hubs.

- **Seafloor Edge-AI and Optical LiDAR:** *If* multi-PMT DOMs are upgraded with radiation-tolerant Edge-AI FPGA chips (not gone into deep research) and solid-state Silicon Photomultipliers (SiPMs), *then* deep-sea submersibles may gain real-time optical LiDAR capability for millimeter-precision 3D mapping of the ocean floor and underwater defense surveillance.

- **Anti-coating Nanotechnology:** *If* novel non-toxic hydrophobic and photocatalytic surface coatings are engineered to prevent biofilm formation on deep-sea glass spheres, *then* commercial maritime shipping can apply these coatings to commercial hulls, drastically reducing hydrodynamic drag and lowering global shipping carbon emissions.

6 Recent Results and Future Outlook

6.1 Discovery of Candidate KM3-230213A

KM3NeT-ARCA has already demonstrated its discovery capabilities during construction. In February 2023, ARCA recorded a landmark ultra-high-energy neutrino candidate, designated `KM3-230213A`. As displayed in Figure 5, the event triggered over 3,500 PMTs across multiple detector lines, arriving from a near-horizontal trajectory ($\cos\theta \approx 0$). In the event selection plane, `KM3-230213A` sits well above the 1% background contour, completely separated from down-going atmospheric muons.

Reconstruction of `KM3-230213A` places its estimated energy in the tens of PeV regime ($\sim 10^8$ GeV), making it one of the most energetic neutrino events ever recorded. Figure 6 contextualizes this observation against the global high-energy spectrum. The event bridges the energy regime between IceCube's High-Energy Starting Events (HESE) and the cosmogenic neutrino flux predicted by the Greisen-Zatsepin-Kuzmin (GZK) effect.

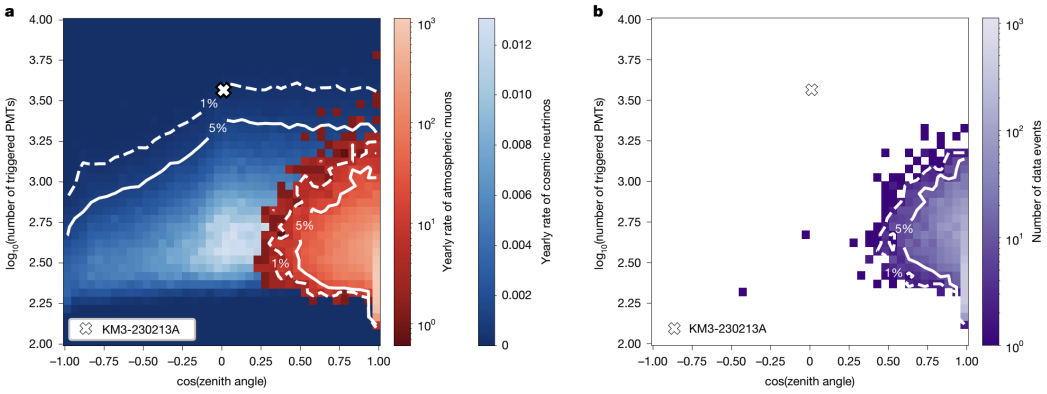


Figure 5: Event selection observable in $\log_{10}(N_{\text{trig}}^{\text{PMT}})$ vs. cos(zenith angle). (a) Simulated atmospheric muon background (red) and cosmic neutrino expectation (blue), showing 1% and 5% contamination contours. (b) Observed data events, featuring the isolated ultra-high-energy candidate `KM3-230213A` (white cross) well above expected backgrounds.

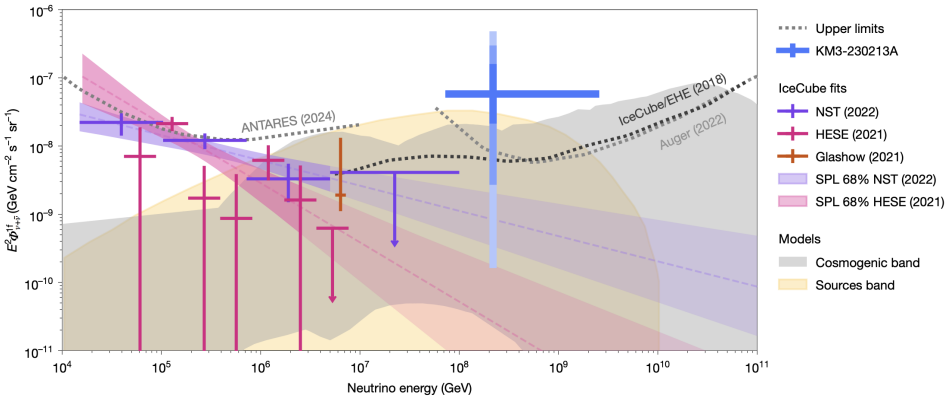


Figure 6: Diffuse neutrino flux ($E^2\Phi$) as a function of neutrino energy spanning 10^4 to 10^{11} GeV. The high-energy candidate `KM3-230213A` (blue cross) is plotted alongside IceCube spectral fits (HESE, Northern Sky Tracks), the Glashow resonance, and cosmogenic neutrino model bands.

6.2 Advanced Countermeasures and the Global Neutrino Network

To overcome the physical volume limits of optical Cherenkov arrays, next-generation observatories will integrate:

1. **Acoustic-Optical Hybrid Detection:** For extreme energies ($E > 10^{17}$ eV), particle cascades produce rapid local thermal expansion, emitting an ultra-short bipolar acoustic pressure pulse. Detecting these thermoacoustic waves with hydrophones permits instrumenting tens of cubic kilometers at low sensor density.

2. **The Global Neutrino Network (GNN):** KM3NeT operates in concert with IceCube-Gen2 (South Pole), Baikal-GVD (Lake Baikal), and the Pacific Ocean Neutrino Experiment (P-ONE off Canada). This distributed array provides 24/7 all-sky coverage, coordinating rapid multi-messenger alerts with gravitational-wave interferometers (LIGO/Virgo/KAGRA) and electromagnetic observatories worldwide.

7 Conclusion

Deep-sea neutrino telescopes represent a major technical and observational significance in the field of astrophysics. From DUMAND-1 to AMANDA and then finally ANTARES and KM3Net was

a complete journey of fails in the making a wonderful device that hence be used for discovering the undiscovered. All the efforts proves that the complex electro-optical instrumentation in the deep sea and glacial region could not only survive but also yield precessional physics. KM3NeT took this on another level by not only mapping cosmic accelerators in the Southern skies but also array to determine the Neutrino Mass Ordering.

Operating in deep seawater presents challenges, including natural ^{40}K radioactivity, bioluminescence, and dynamic currents. Yet the low scattering of deep seawater yields pointing accuracy superior to glacial ice. The engineering demanded by deep-sea physics has yielded real-world spin-offs, from the White Rabbit timing protocol to benthic environmental observatories. With landmark detections such as `KM3-230213A` and the expansion of the Global Neutrino Network, deep-sea neutrino telescopes are poised to play a central role in multimessenger astronomy.

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